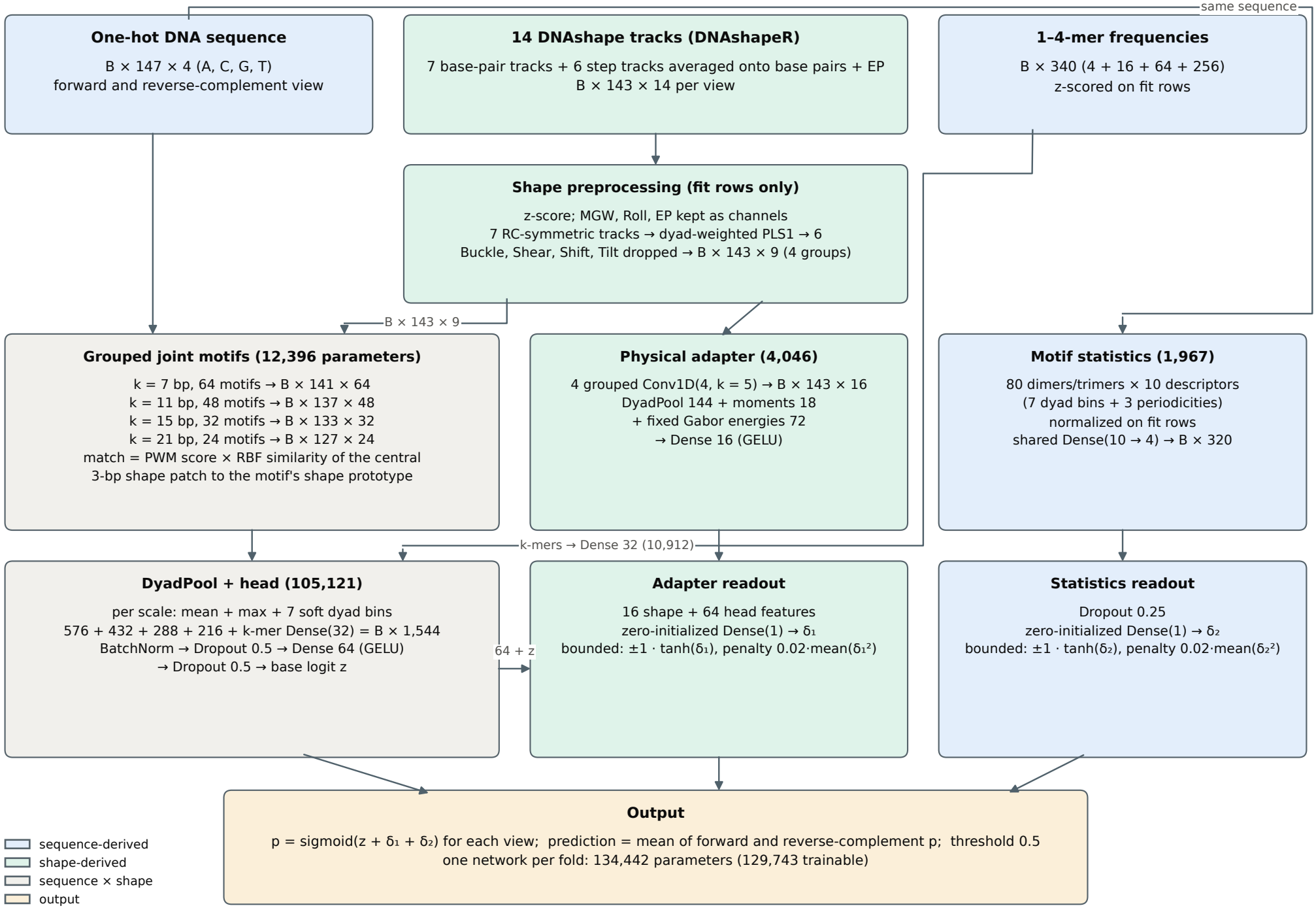
same sequence
One-hot DNA sequence
B × 147 × 4 (A, C, G, T)
forward and reverse-complement view
14 DNAshape tracks (DNAshapeR)
7 base-pair tracks + 6 step tracks averaged onto base pairs + EP
B × 143 × 14 per view
1–4-mer frequencies
B × 340 (4 + 16 + 64 + 256)
z-scored on fit rows
Shape preprocessing (fit rows only)
z-score; MGW, Roll, EP kept as channels
7 RC-symmetric tracks → dyad-weighted PLS1 → 6
Buckle, Shear, Shift, Tilt dropped → B × 143 × 9 (4 groups)
B × 143 × 9
Grouped joint motifs (12,396 parameters)
k = 7 bp, 64 motifs → B × 141 × 64
k = 11 bp, 48 motifs → B × 137 × 48
k = 15 bp, 32 motifs → B × 133 × 32
k = 21 bp, 24 motifs → B × 127 × 24
match = PWM score × RBF similarity of the central
3-bp shape patch to the motif's shape prototype
Physical adapter (4,046)
4 grouped Conv1D(4, k = 5) → B × 143 × 16
DyadPool 144 + moments 18
+ fixed Gabor energies 72
→ Dense 16 (GELU)
Motif statistics (1,967)
80 dimers/trimers × 10 descriptors
(7 dyad bins + 3 periodicities)
normalized on fit rows
shared Dense(10 → 4) → B × 320
k-mers → Dense 32 (10,912)
DyadPool + head (105,121)
per scale: mean + max + 7 soft dyad bins
576 + 432 + 288 + 216 + k-mer Dense(32) = B × 1,544
BatchNorm → Dropout 0.5 → Dense 64 (GELU)
→ Dropout 0.5 → base logit z
64 + z
Adapter readout
16 shape + 64 head features
zero-initialized Dense(1) → δ1
bounded: ±1 · tanh(δ1), penalty 0.02·mean(δ1²)
Statistics readout
Dropout 0.25
zero-initialized Dense(1) → δ2
bounded: ±1 · tanh(δ2), penalty 0.02·mean(δ2²)
Output
p = sigmoid(z + δ1 + δ2) for each view; prediction = mean of forward and reverse-complement p; threshold 0.5
one network per fold: 134,442 parameters (129,743 trainable)
sequence-derived
shape-derived
sequence × shape
output